SerpentGuard AI: Models, Baselines & Pipeline

Offline Indian snake identification from photos or live video: dangerous / not on the WHO list, still keep away / we don't know

CANDIDATE SUBSET	FINAL MODEL (H1_TOP1)	SAFETY VERDICT (FINAL)	ON THE LAPTOP (X1)
13,925	**0.367**	**94.8%**	**13.5 MB**
PHOTOS · 16.4% OF TRAIN	**HEAD MACRO-F1, TUNED, BOX CROP**	**WHO-LISTED CALLED DANGEROUS**	**PACKAGE WITH BM25**
all 184 classes, 6,576 photographers	all 184 classes; seeds within 0.0014	2.1% called not listed; 31.1% false alarms	11.8 FPS on one thread; fp16 loses nothing

What This Deck Covers

I. SET-UP & EVALUATION

- What M3 Asked For & Our Answers
- System, Pipeline & SnakeID Diagrams
- Candidate Subset (16.4% of Train)
- Metrics Built for Rare Classes
- Baselines B0 to B5

II. ARCHITECTURE CHOICES

- E1 Encoder, E2 Head, E3 & E4 Ablations
- Noise Floor & Decision Layer
- Front Ends S1, D1 & the FE Ablation
- Tuning H1 & the Final Model
- Head-to-Head With the Published Model

III. SAFETY, DEPLOYMENT & NEXT

- Decision Rules & the False-Alarm Trade-Off
- Video V1 & Robustness R1
- Retrieval Q1 & Size and Speed X1
- End to End P1 & the Video Demo
- Limitations, Milestone 4 & Team

Group 5: Jayesh Bansal (Team Lead) · Vikas · Puneet Bajaj · Manas Rastogi

The TA's Six Milestone 3 Points, and Where We Answer Each

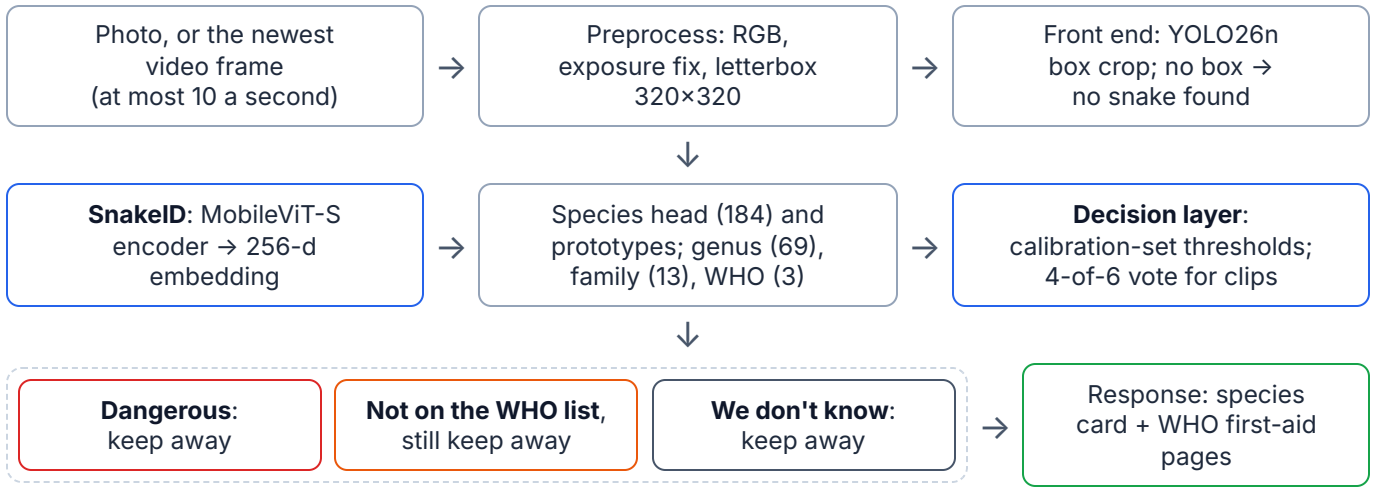
REQUIREMENT	OUR ANSWER	SLIDES
1. Architecture selection	MobileViT-S encoder, ArcFace head, YOLO26n crop, DR0 rules, BM25; each against its alternatives	8, 9, 13, 14, 18, 21
2. Justification	One recipe and a noise floor; the best macro-F1 whose raw WHO recall and O3 are no worse; within noise, the smaller model	8 to 12, 14, 18
3. Baseline, metrics, subset, benchmarks	B0 to B5 and R0; macro-F1 with O2, O3 and false alarms; 13,925-photo subset; head-to-head with B1	5 to 7, 17
4. Hyperparameter tuning	24 random trials over 4 parameters, importances; top 3 retrained with 3 seeds	15, 16
5. End-to-end setup	Photo or clip → verdict, top 3, reason, WHO pages, every node logged; video, robustness, size and speed	3, 19, 20, 22, 23
6. Architecture visualization	Pipeline diagram, labelled SnakeID diagram, demo flow	3, 4, 23

Our Answer in Four Lines

- **Chosen chain**: YOLO26n box crop → MobileViT-S in fp16 with the tuned ArcFace recipe → DR0 verdict rules → BM25 and WHO first-aid pages, all offline.
- **Against the published model**: we cover all 184 classes at 1% of its size; on the 62 species it knows, it still leads (top-1 0.801 against 0.682).
- **Safety trade-off**: 94.8% of WHO-listed snakes called dangerous, just under the 95% target, with 31.1% false alarms; no rule we tried does better.
- **Fits and holds on video**: 13.5 MB with BM25, 11.8 FPS on one laptop thread; 4-of-6 voting keeps 93.0% of clips stable within 2 s.

STATUS	ITEMS	EVIDENCE
Done	Subset, metrics, baselines, E1 to E4, N0, decision layer and rules, S1, D1, FE, H1, retrain, N1, V1, R1, X1, Q1, P1, video demo	Run files in `results/m3/`
PENDING	Each member's own week 3 work-log entry, and every member's signature	Slide 25

One Photo or Frame In, One of Three Verdicts Out, With WHO First-Aid Pages

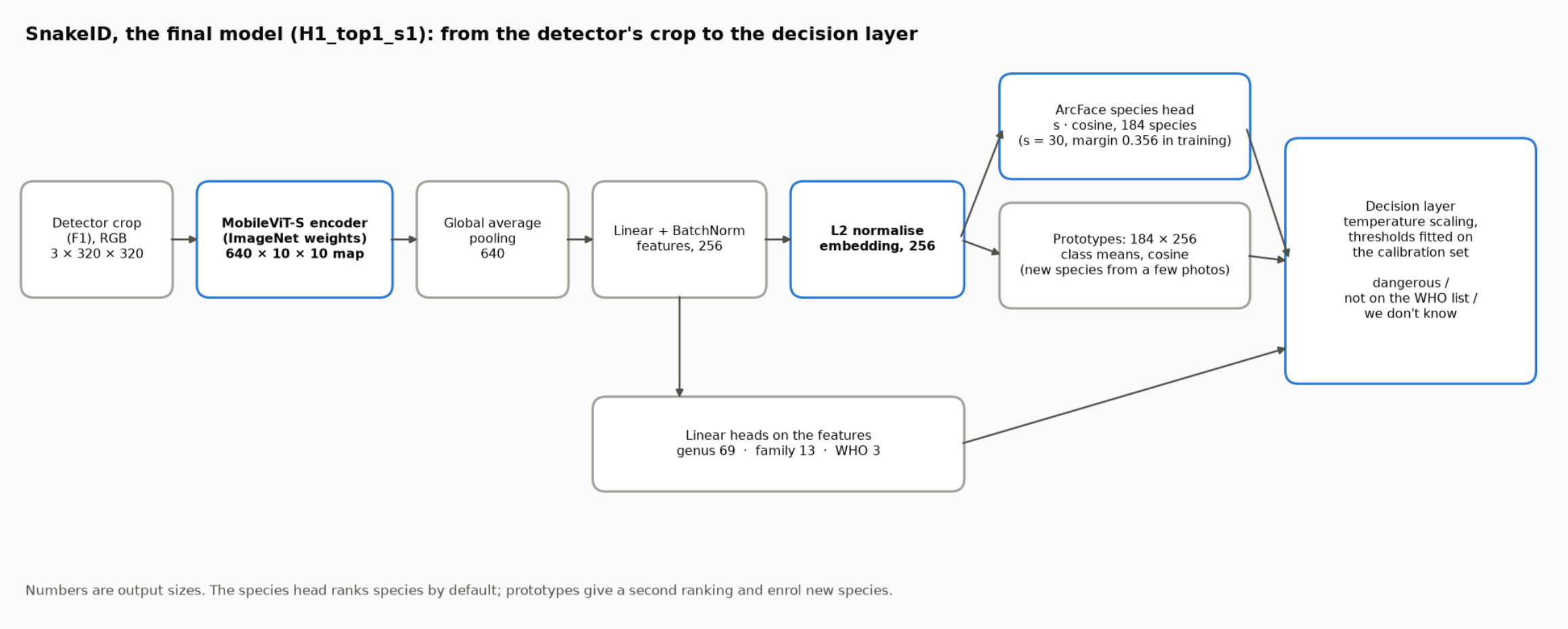


What Each Stage Does

- **Offline and small**: everything runs on the phone, and the app generates no text.
- **SnakeID** names the species by the closest known fingerprint, the way face recognition works; genus and family give a fallback.
- **Dangerous** means the danger table: 50 of 184 classes (6 WHO Category 1, 10 Category 2, 29 more vipers and elapids, 5 keelbacks the guideline names).
- **Response** quotes fixed WHO passages by page; "not listed" is always shown with "still keep away".

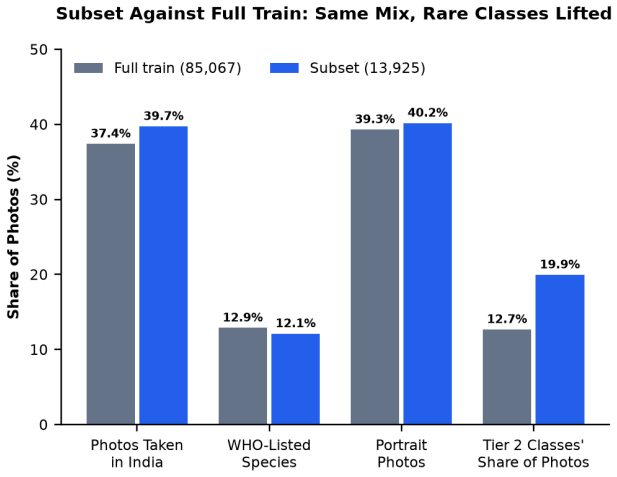
STAGE	INPUT → OUTPUT	CHOSEN (SLIDE)
Front end + SnakeID	photo → crop letterboxed to `(1, 3, 320, 320)` → 256-d embedding; species (184), genus (69), family (13), WHO (3)	YOLO26n box crop (14); MobileViT-S, ArcFace head (8, 9); tuned (16)
Decision + Response	outputs → verdict with a reason → species card and WHO passages with page numbers	Rules fitted on the calibration set (12, 18); BM25, chosen by size (21, 22)

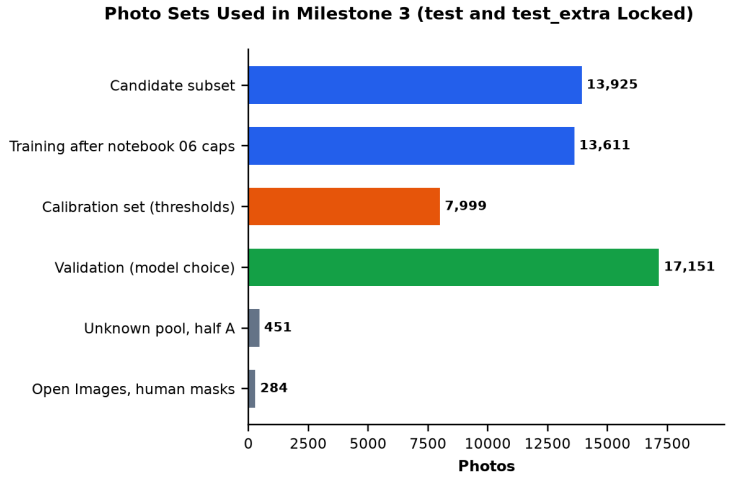
SnakeID: From the Detector's Crop to the Decision Layer



Sizes from the final run's config.

15% of Each Class, at Least 20 Photos: 13,925 Photos (16.4% of Train), All 184 Classes



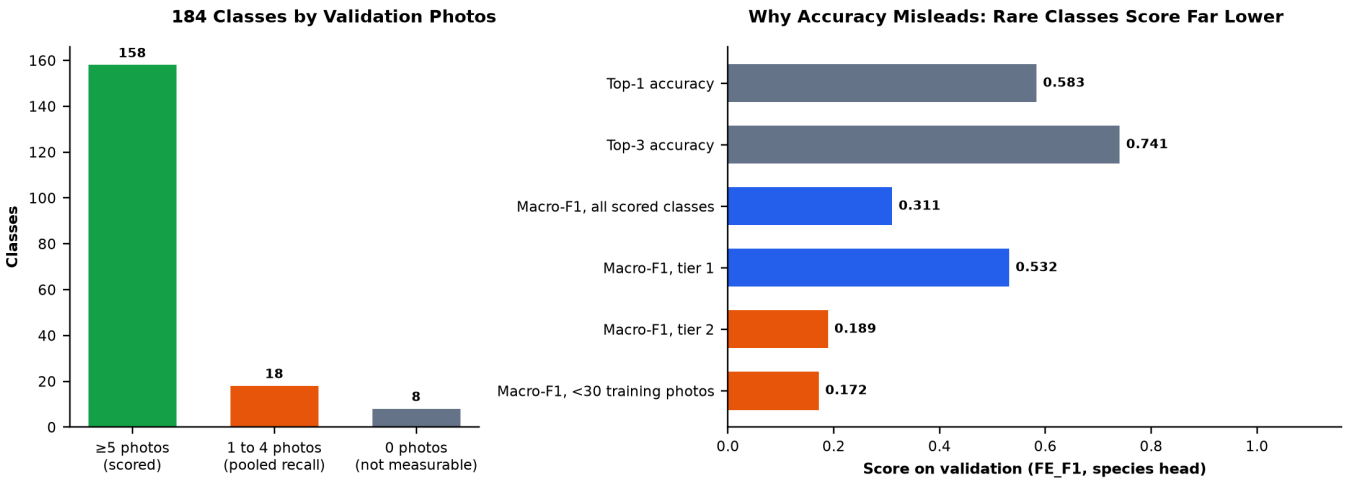


Why It Is Representative

- **Rule**: 15% of each class's training photos, at least 20; photographers take turns so one person can't fill a small class; seed 42. Every run records the subset's SHA-256 (`40196d19a68e`).
- **Same mix as train**: Indian 39.7% vs 37.4%, WHO-listed 12.1% vs 12.9%, portrait 40.2% vs 39.3%; class-mix JS divergence 0.012.
- **Rare classes lifted on purpose**: tier 2 holds 19.9% of the subset against 12.7% of train; 6,576 of 11,272 photographers.
- **Locked**: `test` and `test_extra` are never read by M3 code; a test enforces it.

SET	PHOTOS	USED FOR
Validation	17,151 in 176 classes; 158 have ≥5 photos	Every model choice; 8 classes with none and 18 with 1 to 4 are not measurable, never zero
Calibration	7,999 training photos in 111 classes, outside the subset	Temperatures and verdict thresholds; photographers not in the subset
Unknown pool, half A	451 photos of 49 species we never train on	Unknown detection (U1); half B is kept for Milestone 5

Macro-F1 on the 158 Classes With ≥5 Validation Photos, Guarded by Two Safety Metrics

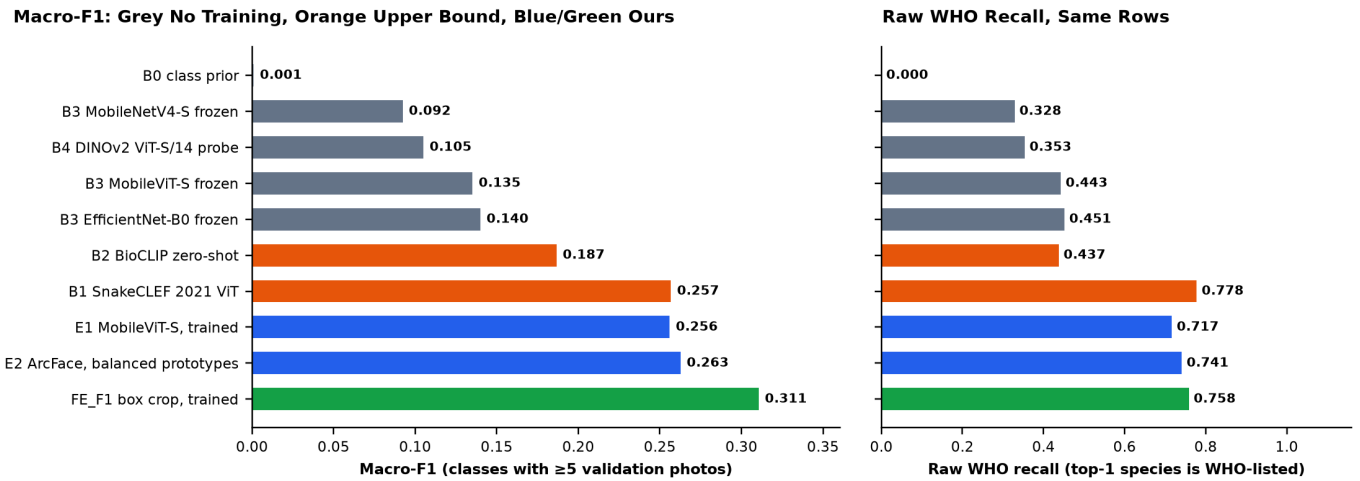


What We Measure & Why

- **Primary: macro-F1 on classes with ≥5 photos**: every class counts equally. Accuracy hides the tail: FE_F1 has 58.3% top-1 but tier 2 macro-F1 of 0.189.
- **O2, WHO recall**: share of WHO-listed photos called dangerous. Raw O2: the top-1 species is WHO-listed.
- **O3**: share of WHO-listed photos called "not listed", the costliest error.
- **False alarms**: snakes outside the WHO list and the viper and elapid families called dangerous. Intervals resample photographers, not photos.

METRIC	DEFINITION	M1 TARGET
O2 · O3	WHO-listed photos called dangerous · called "not listed"	≥ 0.95 · ≤ 2%
Unknown flagged · ECE	Unknown-pool photos called "we don't know" as unfamiliar · calibration error, 15 bins, before and after temperature	≥ 70% · ≤ 0.05

Untrained Baselines Reach 0.001 to 0.257 Macro-F1; Training on 16.4% Lifts MobileViT-S From 0.135 to 0.256

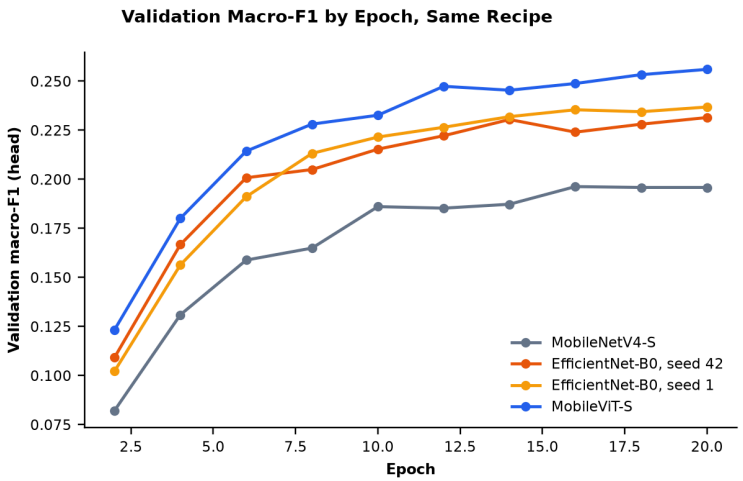


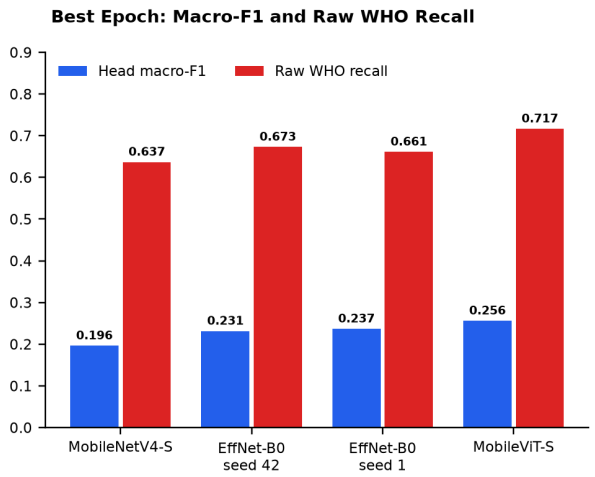
Reading the Baselines

- **B1 is an upper bound**: the published SnakeCLEF 2021 ViT knows 62 of our 184 classes (85.2% of validation photos); on photos uploaded from 2022, after its data, 0.252.
- **B2 BioCLIP zero-shot** (0.187) may have trained on our photos, so it is an upper bound too.
- **Frozen encoders** with nearest prototypes score 0.092 to 0.140; frozen DINOv2 ViT-S/14 with a linear probe 0.105.
- **B5**: yolov8n-oiv7 off the shelf finds few snakes: recall 0.22, mAP@0.5 0.51.

BASELINE (NO TRAINING ON OUR PHOTOS)	MACRO-F1	TOP-1	WHO RECALL (RAW)
B1 SnakeCLEF 2021 ViT, labels mapped to ours	0.257	68.3%	0.778
B2 BioCLIP zero-shot with species names	0.187	35.4%	0.437
B3 EfficientNet-B0 / MobileViT-S frozen, prototypes	0.140 / 0.135	22.9% / 21.7%	0.451 / 0.443

MobileViT-S Wins E1 With 0.256 Head Macro-F1, Ahead of EfficientNet-B0 (0.231, 0.237) and MobileNetV4-S (0.196)





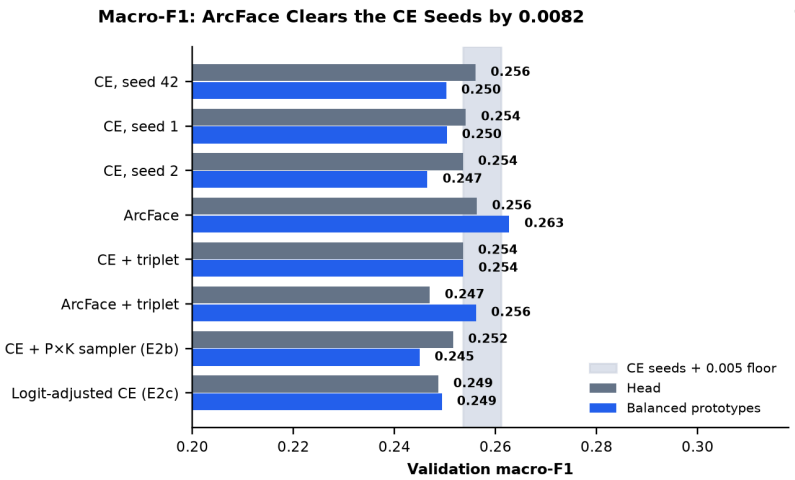
Why MobileViT-S

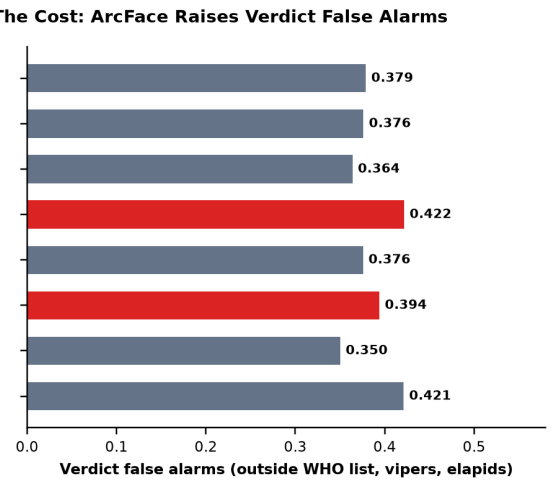
- **One recipe**: AdamW 3e-4 (encoder 0.1×), weight decay 0.05, 20 epochs, batch 64, 320 px, notebook 06's augmentation and sampler.
- **Beyond noise**: EfficientNet-B0's two seeds differ by 0.0054; MobileViT-S leads their mean by 0.022.
- **Best raw WHO recall** (0.717); attention sees the whole snake at once, which suits long thin bodies.
- **Trade-off**: the largest of the three; full int8 breaks it, so it ships in fp16 (X1, slide 22).

ENCODER	HEAD MACRO-F1	TOP-1 / TOP-3	TIER 1 / TIER 2	WHO RECALL (RAW)	GPU WAITING
MobileNetV4-S	0.196	45.4% / 64.3%	0.385 / 0.093	0.637	86%
EfficientNet-B0, seed 42	0.231	49.2% / 67.7%	0.430 / 0.123	0.673	62%
EfficientNet-B0, seed 1	0.237	49.3% / 68.1%	0.431 / 0.130	0.661	56%
MobileViT-S	0.256	53.4% / 71.9%	0.479 / 0.134	0.717	17%

E1 Encoder Comparison on the Candidate Subset

ArcFace Wins E2 by 0.0082, Beyond the 0.005 Floor, but Raises Verdict False Alarms to 42.2%



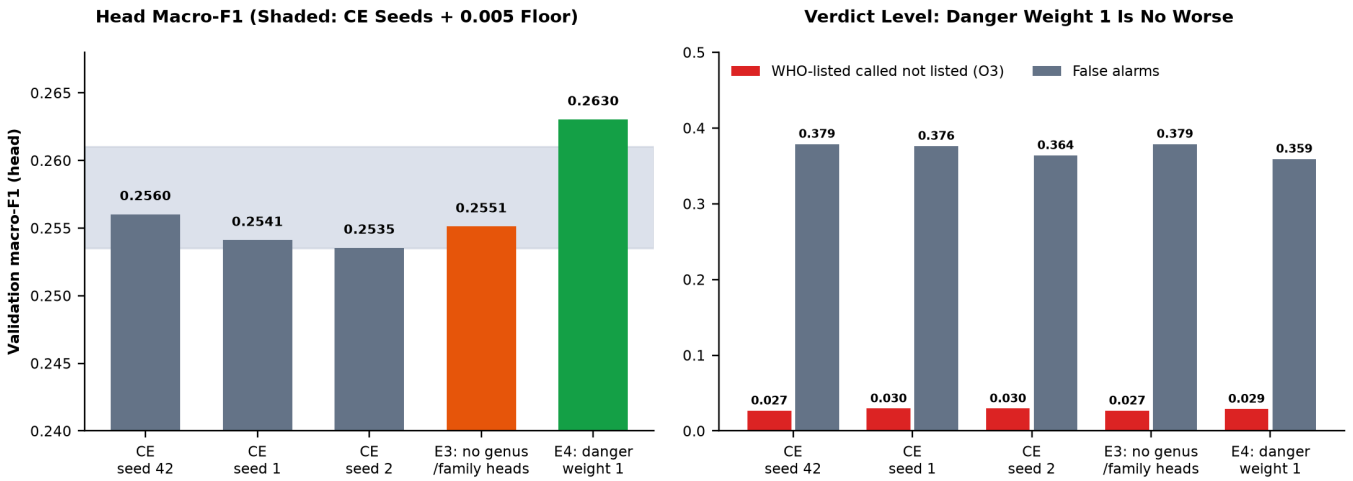


Gains, Costs & Negative Results

- **ArcFace**, balanced prototypes: 0.263 against 0.2546 for CE (head, mean of 3 seeds); raw WHO recall 0.741 vs 0.716, raw O3 0.259 vs 0.284.
- **The cost**: ArcFace's verdicts raise false alarms to 0.422, against 0.364 to 0.379 for the CE seeds.
- **Triplet loss added nothing**: CE + triplet 0.254 at best; ArcFace + triplet 0.256 against 0.263. The P×K sampler alone (E2b): 0.252.
- **Logit-adjusted CE** (E2c): 0.249, and top-1 falls to 37.3%.

RUN (MOBILEVIT-S)	HEAD / PROTOTYPES / BALANCED	TOP-1	WHO RECALL (RAW)	FALSE ALARMS
Cross-entropy (CE), seed 42	0.256 / 0.254 / 0.250	53.4%	0.717	0.379
ArcFace (s = 30, m = 0.3)	0.256 / 0.262 / 0.263	52.9%	0.741	0.422
CE + batch-hard triplet	0.254 / 0.254 / 0.254	53.4%	0.725	0.376
ArcFace + triplet	0.247 / 0.257 / 0.256	51.9%	0.726	0.394

Genus and Family Heads Stay for the Fallback; the 5× Danger Weight Bought Nothing

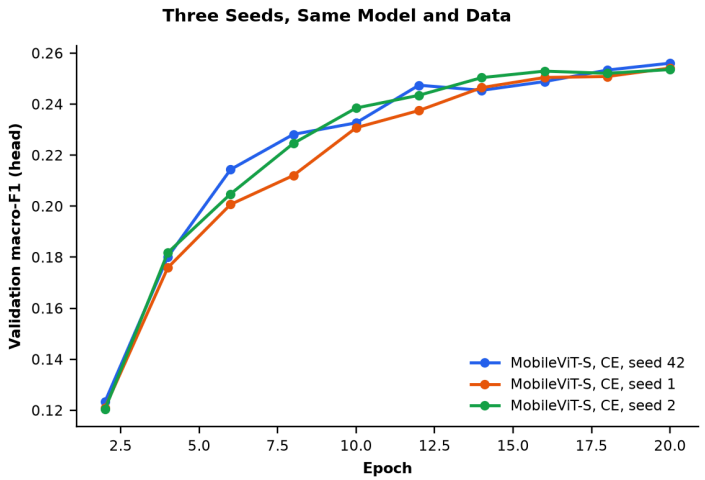


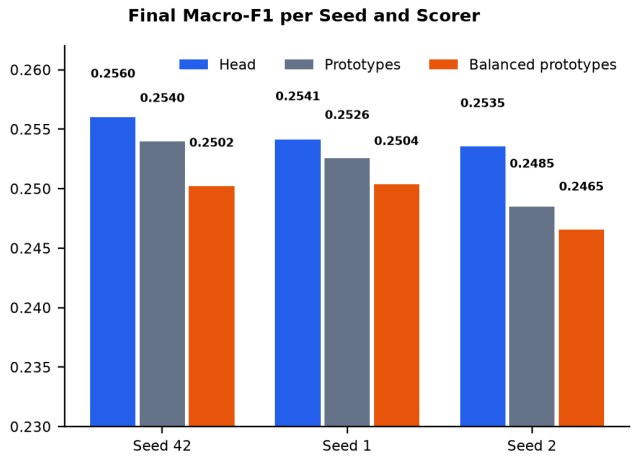
What E3 and E4 Show

- **E3**: dropping the genus and family heads moves head macro-F1 from 0.256 to 0.255, inside the noise. They stay: the fallback needs them.
- **Fallback**: on "we don't know" photos, E2 shows a genus for 17.5% (right 67.8%) and a family for 86.7% (right 79.5%). Summed species probabilities beat the heads: genus 0.612 vs 0.591.
- **E4, weight 5× → 1×**: head 0.263 against 0.2546; verdict O3 0.029 (seeds 0.027 to 0.030), false alarms 0.359 (seeds 0.364 to 0.379). The weight bought nothing and is dropped.
- **WHO head switched off** by the joint threshold fit in 7 of 11 runs: the species head alone reached the recall target with fewer false alarms.

RUN	CHANGE FROM E1'S MOBILEVIT-S	HEAD MACRO-F1	O3 (VERDICT)	FALSE ALARMS
E3	no genus and family heads	0.255	0.027	0.379
E4	danger weight 1 on the WHO head instead of 5	0.263	0.029	0.359

Three Seeds of One Model Span 0.2535 to 0.2560, So a Gap Under 0.005 Counts as No Difference



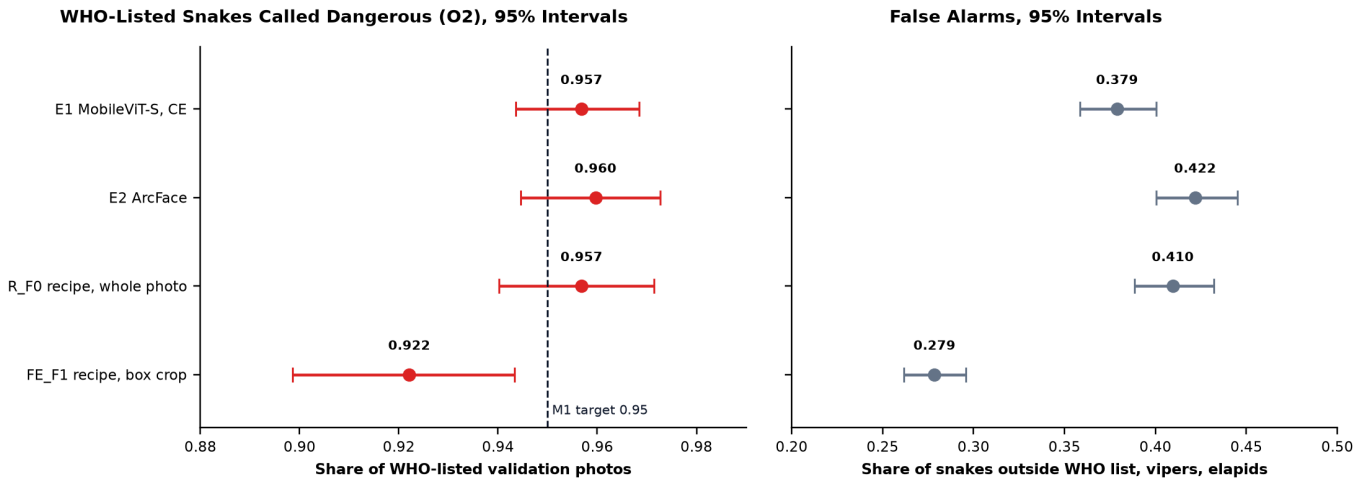


How We Read Every Comparison

- **Spread 0.0025** in head macro-F1 over seeds 42, 1 and 2; twice that, 0.005, is the smallest gap we call real.
- **Clears it**: E1 MobileViT-S (+0.022), E2 ArcFace (+0.0082), FE box crop (+0.053).
- **Doesn't**: E3 (-0.001), triplet loss, the P×K sampler, F1m against F1 (0.002).
- **N1** (slide 16): the final recipe's seeds span 0.0014, so the floor holds for it; H1_top2's span 0.0137, so not for every recipe.

SEED	HEAD	PROTOTYPES	BALANCED	TOP-1	WHO RECALL (RAW)
Seed 42 (E1_mvits)	0.2560	0.2540	0.2502	53.4%	0.717
Seed 1 (N0_s1)	0.2541	0.2526	0.2504	53.1%	0.714
Seed 2 (N0_s2)	0.2535	0.2485	0.2465	53.3%	0.717

Rules Turn a 25.9% Species Error on WHO-Listed Snakes Into 2.7% Wrong Verdicts; the Price Is False Alarms

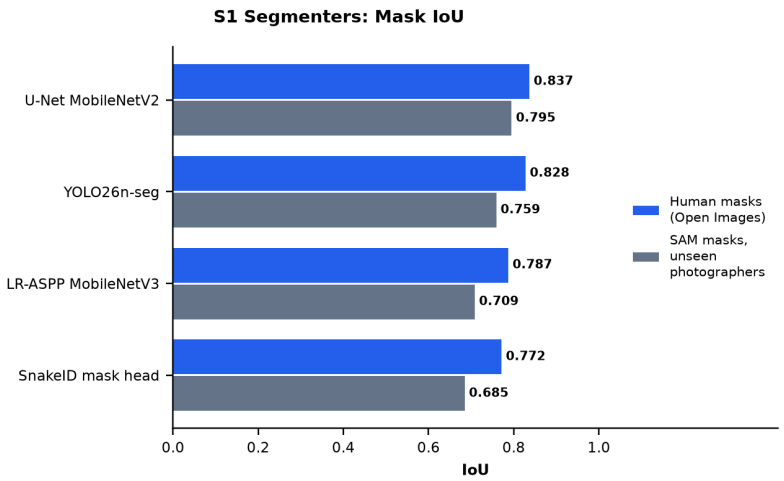


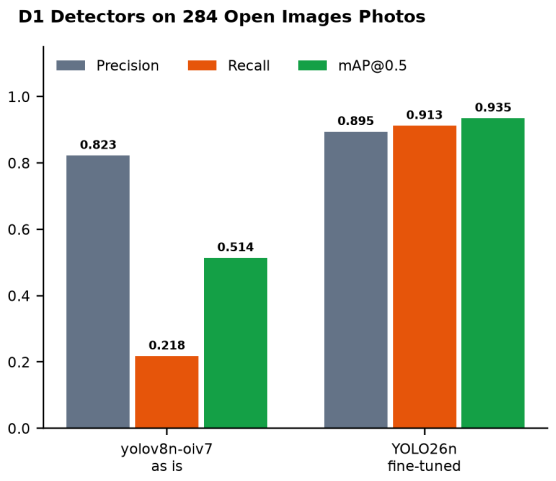
How a Verdict Is Made

- **Dangerous** if WHO-listed species mass reaches θ_d (fitted at 0.05), the top species is in the danger table, or vipers and elapids together reach 0.5.
- **We don't know** if the top species is unsure or far from every prototype; never "not listed" for a genus with a WHO-listed species.
- **Thresholds from the calibration set** (566 WHO-listed photos), never validation; 73 classes have no calibration photos.
- **Calibration**: ArcFace ECE 0.193 before temperature scaling, 0.036 after; error at 80% coverage 0.380, far from 5%.

ON VALIDATION (17,151 PHOTOS)	E1 MOBILEVIT-S, CE	E2 ARCFACE	M1 TARGET
WHO-listed called dangerous (O2)	0.957 (0.944 to 0.968)	0.960 (0.945 to 0.973)	≥ 0.95
WHO-listed called "not listed" (O3)	0.027	0.027	≤ 2%
False alarms	0.379 (0.358 to 0.400)	0.422 (0.400 to 0.445)	
"We don't know"	7.6%	6.6%	

U-Net Segments Best (IoU 0.837 on Human Masks); Fine-Tuned YOLO26n Finds Snakes (mAP@0.5 0.935)



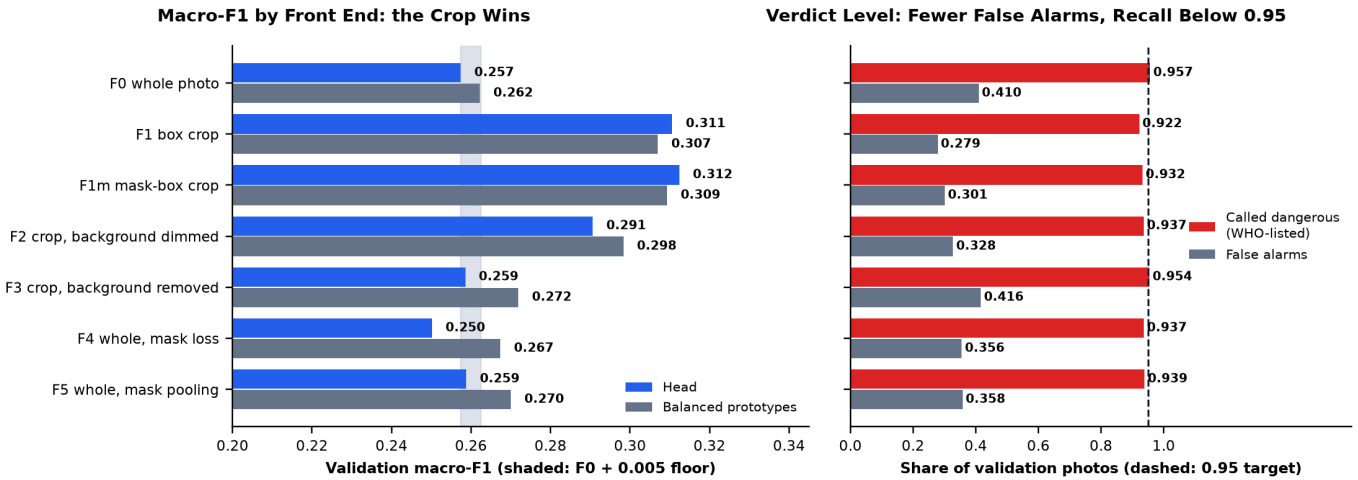


Front-End Candidates

- **Scored on people's labels**: 284 Open Images photos with human-drawn masks and boxes (CC BY 2.0), never our own training labels.
- **S1**: U-Net (6.6M parameters) beats YOLO26n-seg (3.1M) by 0.009, outside the 0.005 tie band.
- **D1**: fine-tuning YOLO26n on the Grounding DINO boxes lifts recall from 0.218 to 0.913, against yolov8n-oiv7 as is.
- **Validation coverage**: 99.95% of photos get a box above confidence 0.05, 91.9% above 0.25; 93.2% a usable mask.

SEGMENTER (S1)	PARAMETERS	IOU, HUMAN MASKS	DICE	IOU, SAM MASKS, UNSEEN
U-Net MobileNetV2	6.6M	0.837	0.902	0.795
YOLO26n-seg	3.1M	0.828	0.896	0.759
LR-ASPP MobileNetV3	3.2M	0.787	0.867	0.709
Mask head on SnakeID's MobileNetV4-S	1.4M	0.772	0.859	0.685

Cropping to the Snake Lifts Head Macro-F1 From 0.257 to 0.311; Segmentation Does Not Beat the Crop

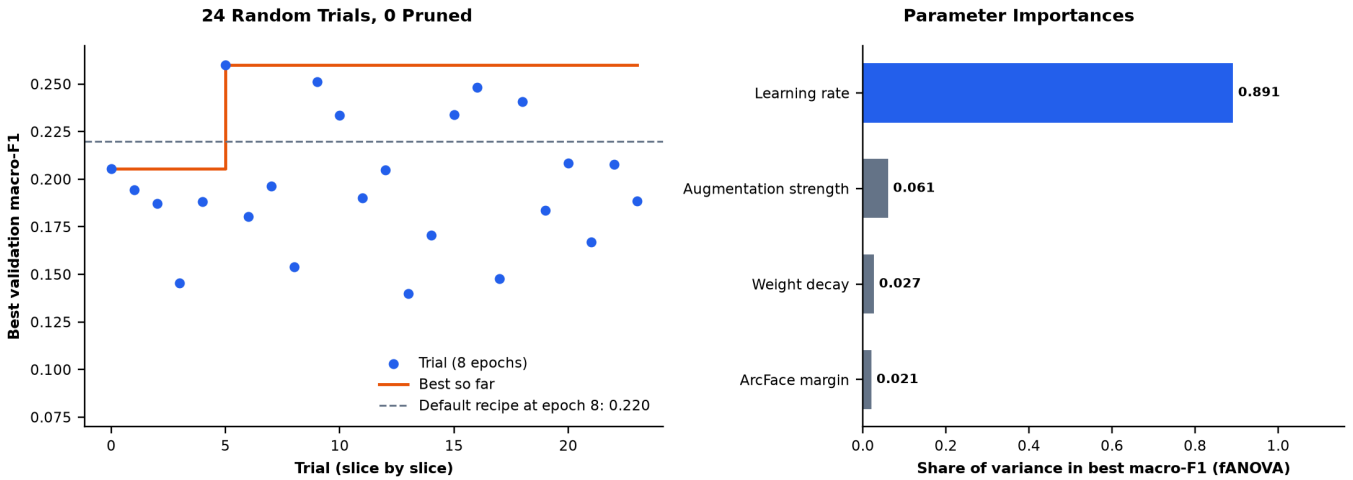


What the Ablation Shows

- **F1 wins**: +0.053 head macro-F1, about ten times the floor. F1m ties (0.002 apart) but needs the 6.6M U-Net, so the U-Net leaves the pipeline.
- **Segmentation does not beat a crop**: dimming the background costs 0.020, removing it 0.052; a mask loss (0.250) or mask pooling (0.259) leaves the whole photo where it was.
- **The verdict moved the wrong way**: on F1, WHO-listed called dangerous falls to 0.922 (0.899 to 0.943), below 0.95, as false alarms fall from 0.410 to 0.279.
- **No box, no snake**: 151 validation photos fall below τ_snake = 0.102 and are scored "no snake"; slide 16 reports the final model's verdicts.

FRONT END (SAME RECIPE)	HEAD / BALANCED	TOP-1	O2 (VERDICT)	FALSE ALARMS
F0: whole photo	0.257 / 0.262	53.1%	0.957	0.410
F1: crop to the YOLO26n box	0.311 / 0.307	58.3%	0.922	0.279
F1m: crop to the predicted mask's box	0.312 / 0.309	58.3%	0.932	0.301
F2 / F3: crop, background dimmed / removed	0.291 / 0.259	56.3% / 53.2%	0.937 / 0.954	0.328 / 0.416

24 Random Trials Over 4 Parameters: the Learning Rate Explains 89% of the Variance

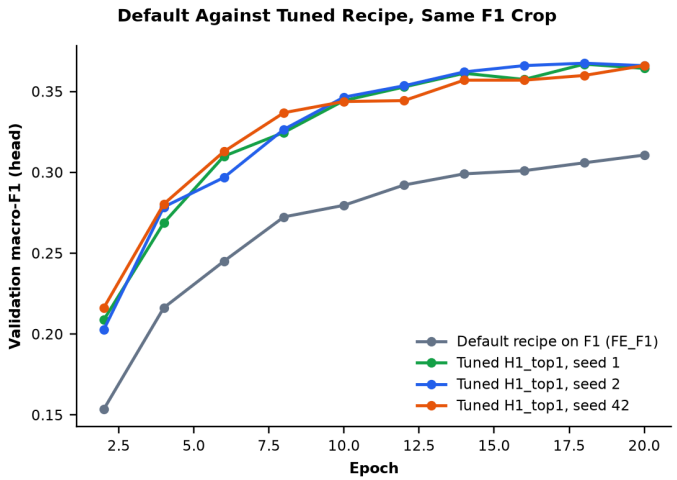


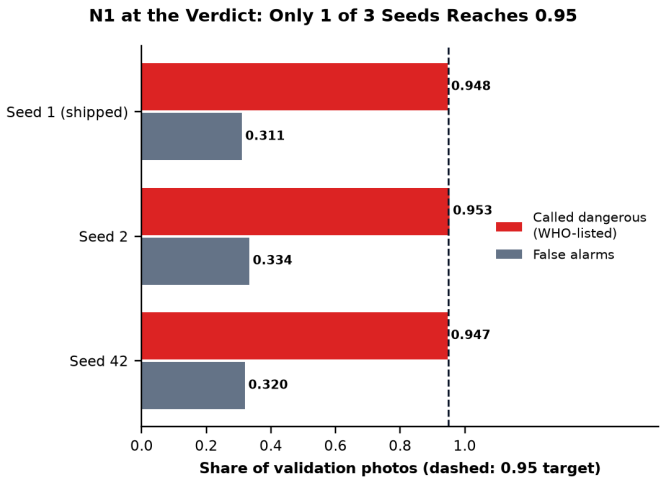
Search, Result & Assumption

- **Space**: learning rate 1e-4 to 1e-3 and weight decay 1e-4 to 0.1 (log scales), augmentation strength 0.5 to 1.5, ArcFace margin 0.1 to 0.5; input stays 320 px.
- **Random search** in 6 slices of 4 trials on three Kaggle accounts, merged into one Optuna study; pruning off, so every trial is a clean sample; 15.7 GPU-hours on T4s.
- **Best: 0.260** at epoch 8, against 0.220 for the default recipe at the same epoch. Not a fair race: the default's learning rate was still high on its 20-epoch schedule.
- **Assumption**: the search ran on the whole photo while FE ran; all 24 trials were still improving at their last epoch. The retrain carries the top 3 to the F1 crop (slide 16).

SETTING	LEARNING RATE	WEIGHT DECAY	AUGMENTATION	MARGIN	MACRO-F1
Default recipe (R_F0)	3e-4	0.05	1.0	0.3	0.220 at epoch 8 of 20
Trial ranked 1	9.5e-4	1.6e-3	0.75	0.356	0.260 at epoch 8
Trial ranked 2	9.4e-4	1.5e-2	0.91	0.346	0.251 at epoch 8
Trial ranked 3	5.8e-4	4.6e-3	0.56	0.391	0.248 at epoch 8

Final Model: Tuning Lifts Head Macro-F1 From 0.311 to 0.367 on the Crop, and Three Seeds Agree Within 0.0014



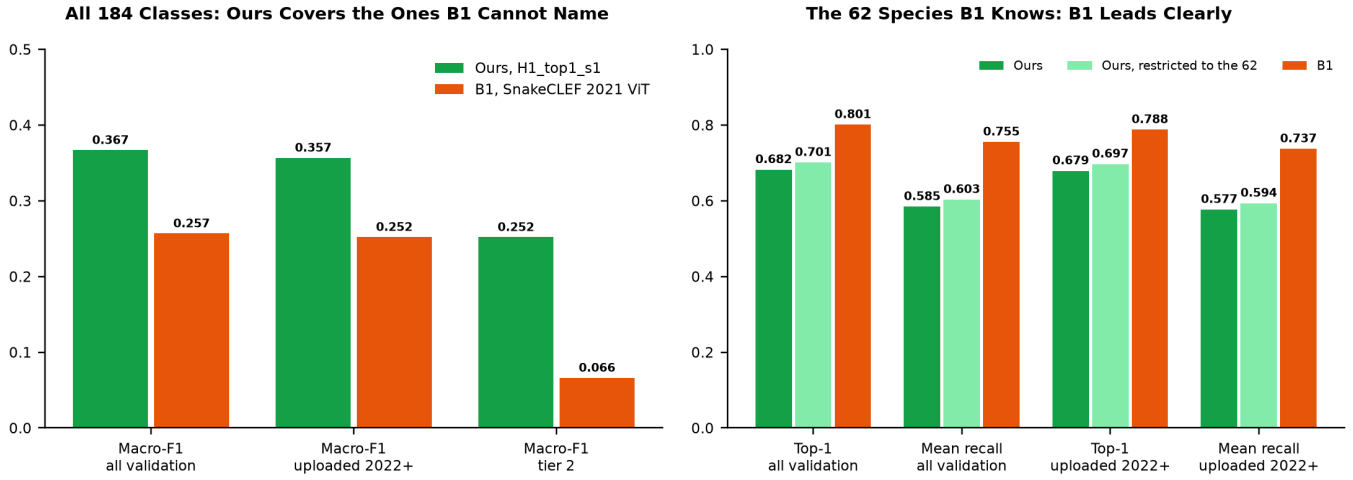


What Ships & How It Scores

- **Recipe**: learning rate 9.5e-4, weight decay 1.6e-3, augmentation 0.75, ArcFace margin 0.356, 20 epochs on the F1 crop. We ship seed 1, the median seed, scored with the head.
- **Gain**: head macro-F1 0.3668 over 3 seeds against 0.311, +0.056; on tier 2 classes 0.189 → 0.252.
- **N1**: seeds 1, 2 and 42 span 0.0014 of head macro-F1, under N0's 0.0025.
- **Verdicts**: θ_d fitted so the lower bound of calibration recall reaches 0.95. WHO-listed called dangerous 0.948 (0.930 to 0.963); only 1 of 3 seeds reach 0.95 on validation.

ON VALIDATION	DEFAULT RECIPE ON F1 (FE_F1)	TUNED, SHIPPED (H1_TOP1_S1)
Head macro-F1 · top-1	0.311 · 58.3%	**0.367** · 62.2%
Tier 1 · tier 2 macro-F1	0.532 · 0.189	0.576 · 0.252
O2 · false alarms · O3	0.922 · 0.279 · 0.035 (point fit)	0.948 · 0.311 · 0.021 (lower-bound fit)

We Cover All 184 Species at 1% of the Size; on the 62 Species B1 Knows, It Still Leads

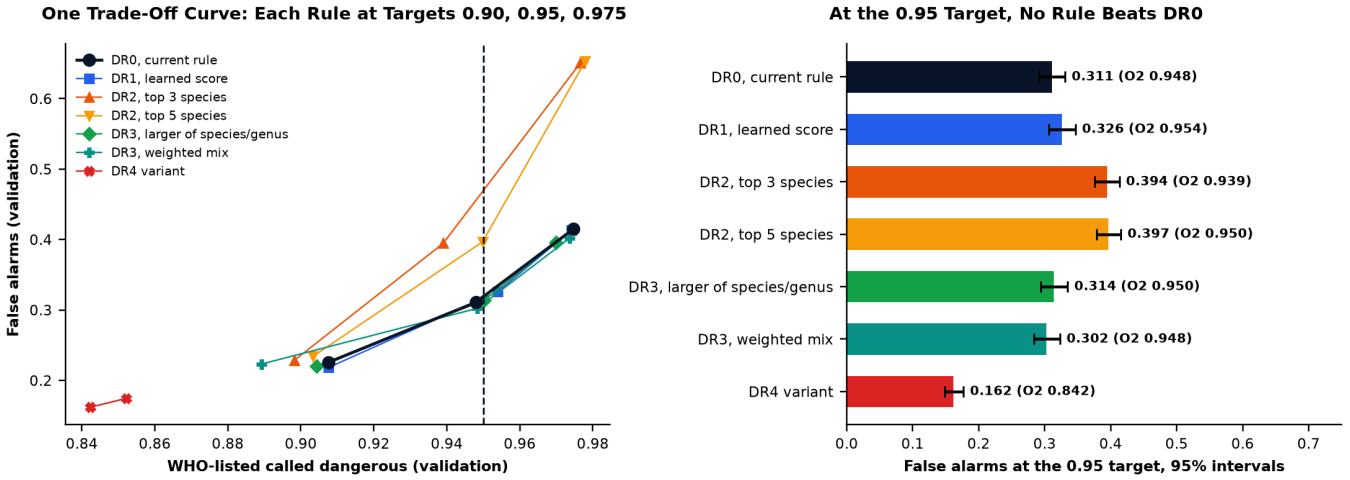


A Fair Test, Read Honestly

- **One scorer for both**: it reproduces our run's macro-F1 exactly, and with shuffled labels ours scores 0.006.
- **Over all 184 classes**: macro-F1 0.367 against 0.257; most of that gap is coverage, since B1 cannot name 122 of our classes (15% of photos).
- **On the 62 species B1 knows, B1 clearly wins**: top-1 0.801 against our 0.682 (0.701 restricted to the 62), mean recall 0.755 against 0.585; from 2022, top-1 0.788 against 0.679.
- **Size**: about 1.13 GB against our 10.7 MB in fp16. Ours trained on 13,611 photos; training on the full set is M4.

ON OUR VALIDATION SET	OURS (H1_TOP1_S1)	B1 (SNAKECLEF 2021 VIT)
All 184 classes: macro-F1 · from 2022	**0.367 · 0.357**	0.257 · 0.252
62 known species: top-1 · mean recall	0.682 · 0.585 (0.701 · 0.603 restricted)	**0.801 · 0.755**
62 known species, from 2022: top-1 · mean recall	0.679 · 0.577	**0.788 · 0.737**
Size	**10.7 MB** in fp16	about 1.13 GB

No Decision Rule Beats DR0: Fewer False Alarms Cost WHO Recall, So the Cut Must Come From the Classifier

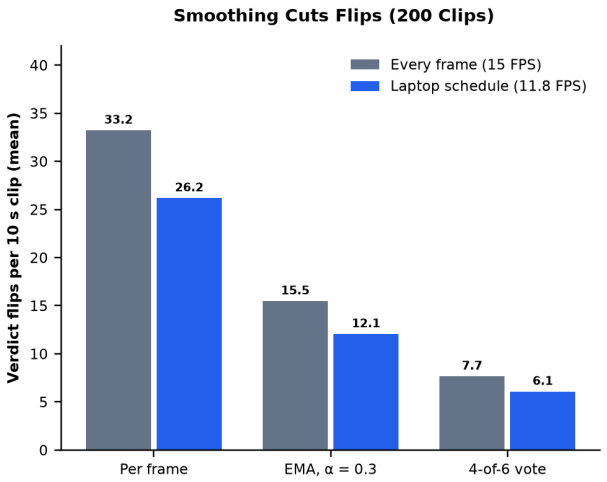


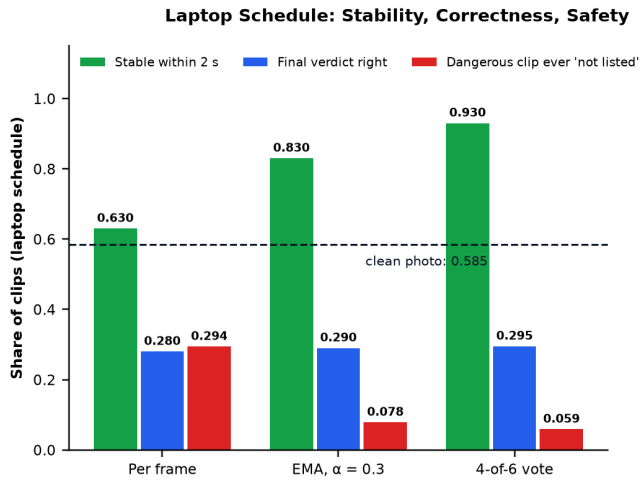
Five Rules, One Curve

- **Fair test**: each rule is fitted on the calibration set at the same recall target and measured once on validation, with the final model.
- **No rule beats DR0**: the weighted mix comes lowest, 0.302 against 0.311, but the intervals overlap; the top-3 and top-5 rules are clearly worse.
- **One curve**: DR0 gives 0.226 false alarms at 0.908 WHO recall, 0.311 at 0.948 and 0.415 at 0.975.
- **The DR4 variant** cuts false alarms to 0.162, but calls only 0.842 of WHO-listed snakes dangerous. A better classifier is the way down: M4 trains on all training photos.

RULE (0.95 TARGET, LOWER-BOUND FIT)	O2	FALSE ALARMS (95% INTERVAL)	O3	WE DON'T KNOW
DR0, current rule	0.948	0.311 (0.292 to 0.331)	0.021	9.5%
DR1, learned score	0.954	0.326 (0.306 to 0.347)	0.016	9.5%
DR3, weighted species and genus mix	0.948	0.302 (0.283 to 0.323)	0.020	10.3%
DR4 variant: weak WHO-only calls become we don't know	0.842	0.162 (0.148 to 0.177)	0.021	22.8%

Video: 4-of-6 Voting Cuts Verdict Flips From 26.2 to 6.1 per Clip and Keeps 93.0% of Clips Stable Within 2 s



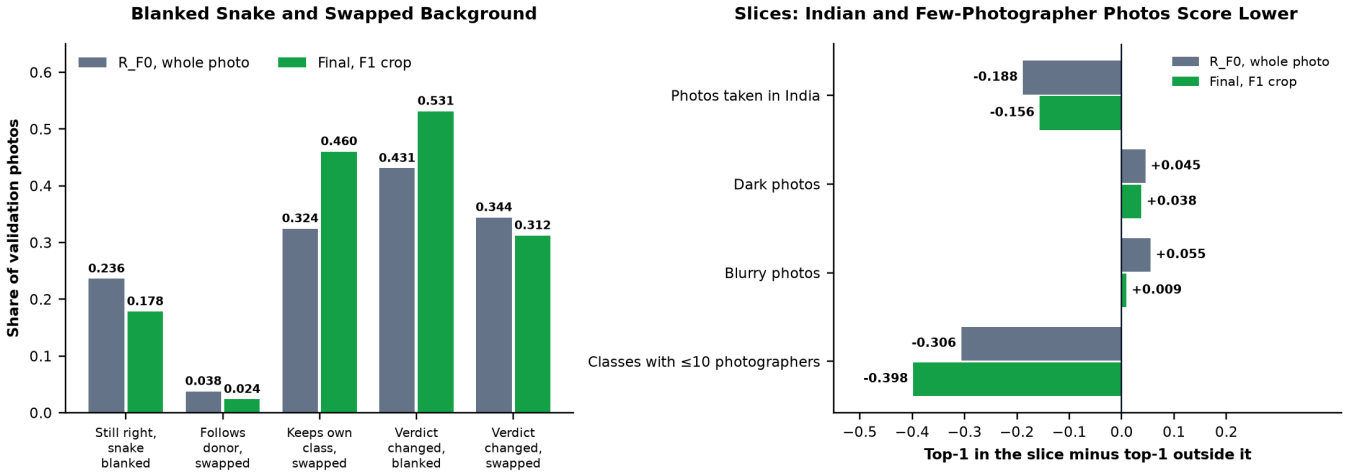


Live Video on Hard Synthetic Clips

- **Set-up**: 200 synthetic 10 s clips at 15 fps from validation photos of 176 classes (camera path, motion blur, low light, noise, JPEG, leaves on the predicted mask); the detector runs on every scored frame.
- **Laptop schedule**: frames scored at the measured 11.8 FPS (84.9 ms a frame), so 21.3% are skipped; 6.2% of scored frames find no snake.
- **Safety**: of 51 clips of danger-table species, 5.9% ever show "not listed" with voting, against 29.4% frame by frame; 2.0% end on it.
- **Hard clips**: the final verdict is right on 29.5% of clips (23.1% to 36.3%), against 58.5% on the clean photos; "we don't know" counts as wrong. Only 40% of clips meet the at-most-1-flip target.

METHOD (LAPTOP SCHEDULE, 11.8 FPS)	FLIPS PER CLIP	STABLE WITHIN 2 S	FINAL VERDICT RIGHT	DANGEROUS EVER "NOT LISTED"
Per frame	26.2	63.0%	28.0%	29.4%
EMA, α = 0.3	12.1	83.0%	29.0%	7.8%
4-of-6 vote	6.1	93.0%	29.5%	5.9%

With the Snake Blanked, 17.8% of Right Answers Stay Right, Against 23.6% on the Whole Photo

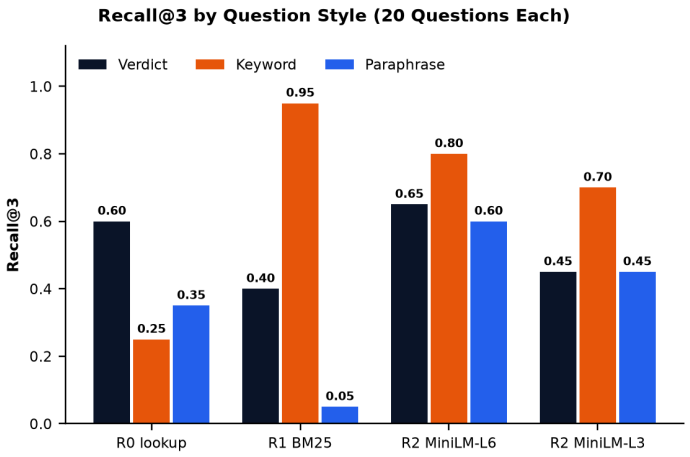


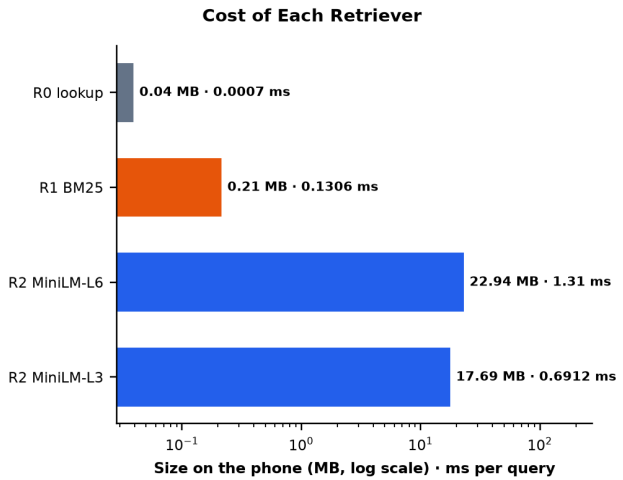
Snake or Scenery?

- **Snake blanked** (grey on the U-Net's predicted mask): of photos the final model gets right, 17.8% stay right (0.163 to 0.194); an upper bound on reliance on scenery. R_F0: 23.6%.
- **Background swapped** with another class's photo: the prediction follows the donor's class for 2.4% (R_F0 3.8%) and keeps its own for 46.0% (32.4%).
- **Verdicts move**: blanking changes 53.1% of verdicts and swapping 31.2%.
- **Weak slices**: top-1 on Indian photos is 15.6 points below the rest (-0.195 to -0.118), and 39.8 points below for classes with ≤10 training photographers.

CONDITION (VALIDATION)	R_F0, WHOLE PHOTO	FINAL, F1 CROP
Still right with the snake blanked	0.236 (0.215 to 0.258)	**0.178** (0.163 to 0.194)
Swapped: follows donor · keeps own class	0.038 · 0.324	0.024 · 0.460
Verdict changed: blanked · swapped	0.431 · 0.344	0.531 · 0.312
Top-1 gap: India · ≤10 photographers	-0.188 · -0.306	-0.156 · -0.398

MiniLM-L6 int8 Finds the Right WHO Passage in the Top 3 for 68% of Questions, Against 47% for BM25



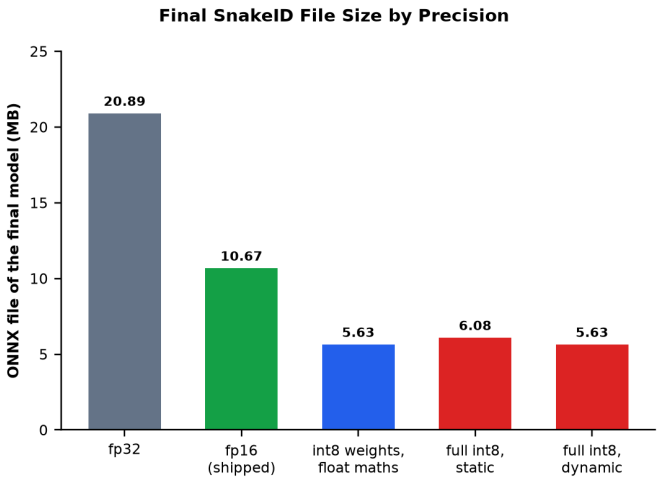


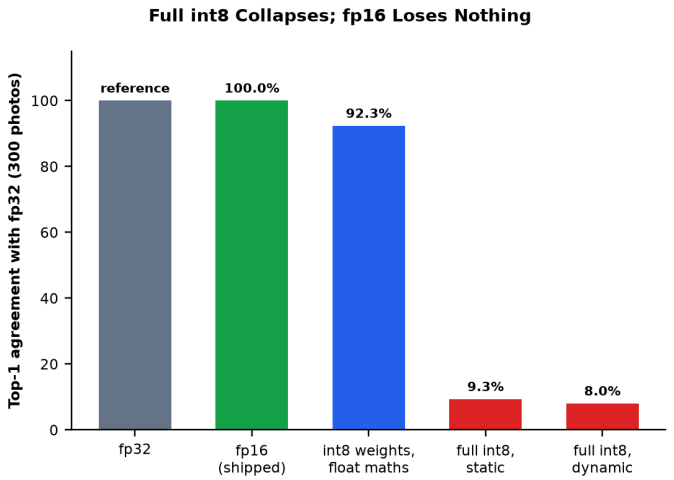
Offline Retrieval, No Generation

- **Knowledge base**: 168 WHO passages cited by page and 184 species cards in our own words; passage text is extracted from the PDF at build time and never committed.
- **60 draft questions**, 20 each of verdict, keyword and paraphrase, for the team to check.
- **MiniLM-L6 beats BM25** on 20 questions and loses on 7 (sign test p = 0.019); the gain is all paraphrases, 11 to 0 (p = 0.001).
- **The M3 package ships BM25**: with MiniLM-L6 it would be 36.24 MB, over the 30 MB budget (slide 22). R0 always shows the first-aid passages.

RETRIEVER	RECALL@1	RECALL@3	MRR	MS PER QUERY	MB
R0 lookup by verdict and species	0.10	0.40	0.265	0.0007	0.04
R1 BM25 (shipped)	0.42	0.47	0.455	0.1306	0.21
R2 MiniLM-L6, int8	0.40	0.68	0.573	1.31	22.94
R2 MiniLM-L3, int8	0.33	0.53	0.462	0.6912	17.69

The Package Fits: 13.5 MB With BM25 and 11.8 FPS on One Laptop Thread; Full int8 Collapses, fp16 Loses Nothing



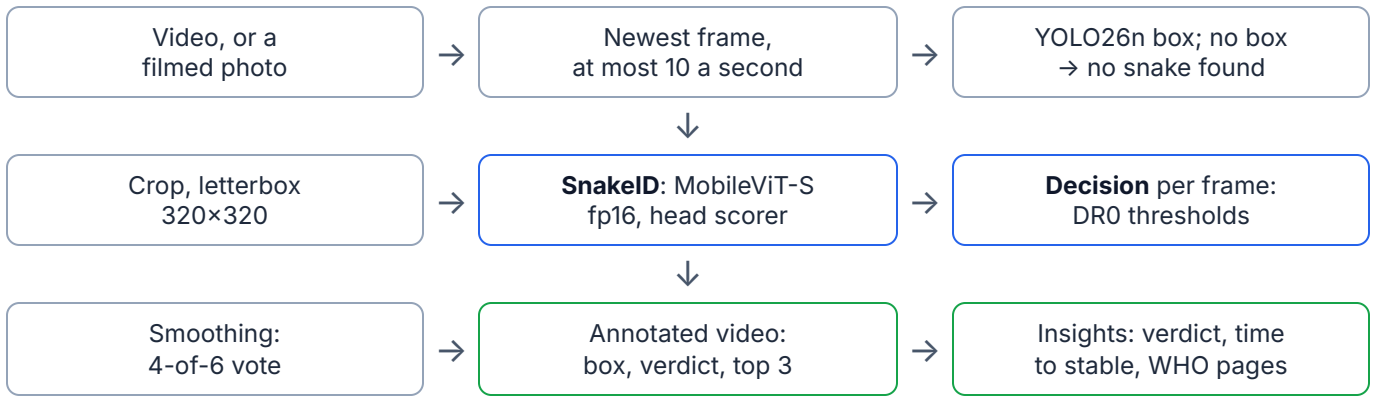


What Ships & Why

- **SnakeID ships in fp16**: 10.67 MB, the same top-1 as fp32 on all 300 photos. int8 weights with float maths: 5.63 MB, 92.3% agreement.
- **Full int8 collapses**: static 9.3% and dynamic 8.0% top-1 agreement; outliers in the trained weights, which untrained exports hid.
- **Detector**: YOLO26n with int8 weights, 2.65 MB, finds fp32's box on 99.7% of photos where fp32 finds one.
- **Package**: 13.51 MB with BM25 against 30 MB; 11.8 FPS on one thread and 21.4 on four, above the 10 FPS target.

NETWORK (LAPTOP CPU, 320 PX)	MB	AGREES WITH FP32	MS, 1 / 4 THREADS
SnakeID, fp16 (shipped)	10.67	100.0%	75.0 / 39.0
SnakeID, int8 weights	5.63	92.3%	58.9 / 24.9
SnakeID, full int8, static	6.08	9.3%	55.3 / 30.7
Detector YOLO26n, int8 weights (shipped)	2.65	99.7%	9.9 / 7.7

End to End: 25 of 26 Dangerous Snakes Caught; the Video Demo Runs at 80 to 88 ms a Frame



P1 Run & the Video Demo

- **P1**: 100 validation photos, one per class, and 5 clips through the detector, the final model and its decision file on a laptop CPU: 87.4 ms a photo.
- **Check**: matches the training run's saved outputs on 98% of verdicts and 98% of species (control 37% and 1%), after a fix: every detector call now pads to the same square, so boxes match training at median IoU 0.999 (0.944 before).
- **Demo**: three CC-licensed photos, each filmed as a 10 s synthetic clip: *Bungarus caeruleus* → "dangerous: WHO Category 1"; *Coelognathus helena* → "not on the WHO list, still keep away"; *Dendrelaphis subocularis* → "we don't know: unclear between species".
- **Every screen** shows the WHO first-aid passages with their page; the demo takes 79.9 to 88.2 ms a frame.

INPUT (FINAL MODEL, LAPTOP)	CALLED DANGEROUS	NOT ON THE WHO LIST	WE DON'T KNOW
26 photos of danger-table species	25	0	1
74 photos outside the table	21	40	13
5 P1 clips	4	0	1
3 demo clips (1 danger-table species)	1	1	1

P1 on the Final Model & the Video Demo Flow (No Frames Shown)

What Does Not Work Yet, and What Milestone 4 Does About It

LIMITATION	EVIDENCE	NEXT
Unknown species	7.5% of unknown-pool photos flagged (target 70%); best AUROC 0.702	Cross-photographer positives, several prototypes per species
False alarms	31.1% of snakes outside the WHO list and the two venomous families; no rule beats DR0	A better classifier: M4 trains on all training photos
WHO recall at the verdict	0.948 (0.930 to 0.963); 1 of 3 seeds reach 0.95	The same; thresholds frozen before any test run in M5
Weak slices	Top-1 15.6 points lower on Indian photos, 39.8 for classes with ≤10 photographers	The full training set in M4
Calibration set	Training photos from other photographers; 73 classes have none	Thresholds frozen before any test run in M5
Video on hard clips	Final verdict right on 29.5% of degraded clips against 58.5% clean	Re-captured real clips, as M1 planned
Full int8	Top-1 agreement 9.3%; fp16 ships	int8 weights as the size fallback

Milestone 4

- **Train the chosen recipe** on the full training set for longer; the default recipe was still rising at its last epoch.
- **Build the Android package** from the shipped files and time it on a phone.
- **Run the 200-photo hand check** promised in Milestone 2.
- **WHO text licence**: shipping passages in the app needs WHO's permission, a team decision.

NEGATIVE RESULT WE REPORT	NUMBER
ArcFace raises false alarms · the danger weight bought nothing	0.422 vs 0.364 to 0.379 · weight 1 scores 0.263, verdicts no worse
Triplet loss added nothing · segmentation doesn't beat a crop	0.256 vs 0.263 · F2 0.291, F3 0.259 vs F1 0.311
No decision rule beats DR0 · full int8 collapses	best rule 0.302 vs 0.311, intervals overlap · 8.0% to 9.3% agreement

What Is Done in Milestone 3, What Is Left, and Group 5 Team

STAGE	STATUS	WHAT IT DELIVERED OR NEEDS
Subset, metrics, baselines, E1 to E4, N0	Done in M3	13,925-photo subset; MobileViT-S, ArcFace, genus and family heads; noise floor 0.005
S1, D1, FE, H1, retrain, N1	Done in M3	YOLO26n box crop; tuned recipe 0.367 head macro-F1, seeds within 0.0014
Decision rules, R1, X1, Q1	Done in M3	DR0 stays; package 13.51 MB with BM25, 11.8 FPS on one thread
V1, P1, video demo	Done in M3	4-of-6 voting: 6.1 flips per clip, 93.0% stable within 2 s; 3 demo clips
Full data, Android package, hand check	Next · M4	Train on the full training set, build the Android package, check 200 photos by hand

Group 5 Project Team · IIT Madras BS in Data Science & AI Lab (2026 T3)

JAYESH BANSAL · TEAM LEAD	VIKAS · TEAM MEMBER	PUNEET BAJAJ · TEAM MEMBER	MANAS RASTOGI · TEAM MEMBER
PENDING: his week 3 entry in the work log **Signed**: PENDING	PENDING: his week 3 entry in the work log **Signed**: PENDING	M3 plan, subset, metrics, model, losses, training runner, baselines, front ends, decision layer, robustness, video, retrieval, export, pipeline, tuning, report draft **Signed**: PENDING	PENDING: his week 3 entry in the work log **Signed**: PENDING